

ReadEase: AI-Powered Dyslexia-Aware Reading Assistant for Malayalam and English

1. Problem Statement

Reading disorders such as dyslexia affect a significant number of school children, making it difficult for them to recognize words, maintain reading fluency, and comprehend text effectively. In many schools, particularly in regional language education, dyslexia often goes undiagnosed due to the lack of accessible screening tools and limited awareness among teachers and parents. As a result, children may experience poor academic performance, reduced confidence, and learning difficulties throughout their education.

Most existing dyslexia support applications are designed primarily for the English language, require expensive proprietary software or specialized eye-tracking hardware, and offer limited support for regional languages such as Malayalam. Consequently, students studying in bilingual educational environments do not receive adequate technological assistance.

ReadEase aims to bridge this gap by providing an AI-powered reading assistant that supports both Malayalam and English. Using a standard webcam, the system estimates the user's gaze and reading behavior through computer vision techniques without requiring dedicated eye-tracking hardware. AI models analyze reading patterns such as prolonged fixation, repeated line reading, and irregular eye movements to identify possible reading difficulties. The platform further assists learners by rendering dyslexia-friendly fonts, highlighting syllables during reading, and providing synchronized text-to-speech assistance to improve reading fluency and comprehension.

Target Users

- Primary and secondary school students
- Children with suspected dyslexia
- Teachers
- Parents
- Special educators
- Educational institutions

Impact if the Problem is Not Solved

- Undiagnosed learning disabilities
- Poor reading fluency

- Reduced academic performance
 - Lower student confidence
 - Limited accessibility for Malayalam learners
 - Delayed intervention for children requiring support
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2. Project Objectives

Main Objective

To develop an AI-powered bilingual reading assistant that detects potential dyslexia-related reading difficulties using webcam-based gaze estimation and provides personalized reading assistance in Malayalam and English.

Specific Objectives

- Develop a webcam-based gaze estimation system using OpenCV and AI models.
 - Detect reading struggle patterns such as excessive fixation, regressions, and skipped words.
 - Support reading in both Malayalam and English.
 - Render text using dyslexia-friendly fonts and layouts.
 - Highlight syllables dynamically during reading.
 - Integrate Text-to-Speech (TTS) for synchronized reading assistance.
 - Generate basic reading performance reports for teachers and parents.
 - Improve reading accessibility and early identification of dyslexia symptoms.
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3. Scope of the Project

In-Scope Features

User Management

- Student registration
- Teacher login
- Reading profile management
- Reading history

Reading Assessment Module

- Webcam-based gaze estimation
- Reading speed calculation
- Fixation detection
- Reading pattern analysis

AI Analysis Module

- Reading difficulty detection
- Basic dyslexia risk estimation
- Reading performance scoring
- AI-generated insights

Reading Assistance Module

- Dyslexia-friendly font rendering
- Adjustable font size
- Syllable highlighting
- Word highlighting

Text-to-Speech Module

- Malayalam speech synthesis
- English speech synthesis
- Synchronized reading
- Adjustable speech speed

Dashboard & Reports

- Reading statistics
- Session history
- Progress tracking
- Teacher performance reports

Out-of-Scope Features

- Clinical diagnosis of dyslexia
- Medical recommendations

- Brain signal analysis (EEG)
 - Professional psychological assessment
 - Mobile application
 - Advanced eye-tracking hardware integration
 - Cloud-based student assessment portal
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4. Proposed Methodology

The project follows the **Agile (Scrum)** software development methodology, with development divided into iterative sprints that deliver functional features incrementally.

Phase 1 – Requirement Analysis

- Study dyslexia-related reading challenges
- Gather educational requirements
- Identify AI and accessibility requirements

Phase 2 – System Design

- UI/UX design
- Database design
- AI model architecture
- Reading workflow design

Phase 3 – Implementation

- User Authentication Module
- Reading Interface
- Webcam Gaze Detection
- AI Reading Analysis
- Text-to-Speech Integration
- Reading Dashboard
- Database Integration

Phase 4 – Testing

- Unit testing

- AI model evaluation
- Integration testing
- Accessibility testing
- User acceptance testing

Phase 5 – Deployment

- Web application deployment
 - AI model integration
 - Final system demonstration
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5. Tools and Technologies

Frontend

- Angular
- TypeScript
- HTML5
- CSS3
- Tailwind CSS

Backend

- Node.js
- Express.js

Database

- PostgreSQL (or MongoDB)

Artificial Intelligence & Computer Vision

- Python
- OpenCV
- MediaPipe Face Mesh
- TensorFlow / PyTorch
- Scikit-learn
- NumPy

Speech Technologies

- Malayalam Text-to-Speech (TTS)
- English Text-to-Speech
- Web Speech API

Development Tools

- Visual Studio Code
 - Git
 - GitHub
 - Postman
 - Jupyter Notebook
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6. Expected Outcome / Deliverables

The project is expected to deliver:

- A bilingual AI-powered reading assistant
 - Webcam-based gaze estimation system
 - Reading struggle detection module
 - Dyslexia-friendly reading interface
 - Malayalam and English Text-to-Speech support
 - AI-generated reading performance reports
 - Teacher dashboard
 - Student progress tracking
 - Complete project documentation
 - Final project presentation and demonstration
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7. Feasibility Study

Technical Feasibility

The project is technically feasible because it utilizes mature open-source technologies such as OpenCV, MediaPipe, TensorFlow, Angular, Node.js, and PostgreSQL. Webcam-

based gaze estimation eliminates the need for expensive eye-tracking hardware, making the solution practical for educational institutions.

Operational Feasibility

The proposed system can be easily deployed in schools, computer laboratories, and educational institutions using standard computers equipped with webcams. Teachers and students require minimal training to operate the system, making it suitable for regular classroom use.

Time Feasibility

The project can be completed within the allotted Mini Project duration by following an Agile development approach. Individual modules such as gaze estimation, reading assistance, and reporting can be developed independently and integrated incrementally.

Economic Feasibility

The project is economically feasible as it uses open-source software frameworks, AI libraries, and standard webcams already available in most educational institutions. No specialized eye-tracking hardware or expensive commercial software is required, significantly reducing implementation costs.

Conclusion

ReadEase is an AI-powered bilingual reading assistant designed to improve reading accessibility for students by combining computer vision, artificial intelligence, and assistive technologies. Through webcam-based gaze estimation, dyslexia-friendly text rendering, syllable highlighting, and synchronized Text-to-Speech support, the system enables early identification of reading difficulties while enhancing reading fluency in both Malayalam and English. By providing an affordable and accessible solution tailored to regional language education, ReadEase contributes to inclusive learning and supports the digital transformation of education, making it a strong and innovative MCA Mini Project.